Aafthab Ali Shaik

Software Engineer

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Professional Summary

Im a final-year Computer Science student who enjoys building things end to end, from the database and APIs to the UI people actually use. My basics in data structures, algorithms and core CS subjects are strong, and Ive used them to ship full-stack and ML projects with React.js, Node.js and that real users can access. Im always curious about new tools and technologies, and once I learn something, I like to put it into practice right away by building with it.

Technical Skills

Programming: C, Python, Java, C Sharp
Frontend: React.js, TypeScript, JavaScript, Vite
Backend: Node.js, Express.js, Flask, FastAPI, REST APIs, .NET
Databases: MySQL, PostgreSQL, MongoDB, Firebase, Supabase
Cloud & Tools: Git, GitHub, GitLab CI/CD, GitLab Runner, Docker, Render, Vercel, Hugging Face Transformers.js
Testing: Pytest, Playwright, Vitest, React Testing Library
Core Concepts: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Static Code Analysis, Embedded Software Build Automation

Experience

•Defence Research and Development Organisation (DRDO) Research Centre Imarat (RCI) *May 2026 – Jul 2026*
Software Engineering Intern Hyderabad

–Set up a GitLab CI/CD pipeline from scratch to build, verify and run static analysis on embedded software.
–Configured GitLab Runners and Docker containers so embedded builds run automatically instead of manually.
–Ran static code analysis and set up auto-generated build reports, which made it easier to catch quality issues early.
–Worked closely with DRDO scientists on embedded software tasks and helped streamline their DevOps workflow.

•Centre for Extended Reality (CXR), GITAM University *May 2026 – Jun 2026*
Full Stack Developer Intern Visakhapatnam

–Worked on the **CXR Web Portfolio**, improving both frontend and backend so the site is responsive and easy to use.
–Integrated Firebase for authentication, data storage, retrieval and real-time updates, and made sure the frontend and backend talked to each other smoothly.
–Cleaned up the codebase by fixing bugs, patching security issues and improving performance, getting it ready for a stable production release.

•Suvehi Interactive Technologies Pvt. Ltd. *Jan 2026 – Apr 2026*
Software Engineering Intern Visakhapatnam

–Built the companys internal website, handling both the responsive frontend and the backend features.
–Contributed to several projects, including website development, game version upgrades, new features, maintenance and bug fixes.

•Infosys Springboard *Dec 2025 – Feb 2026*
Full Stack Developer Intern

–Built a full-stack web app using React.js, Node.js and MySQL.
–Wrote RESTful APIs and improved how the backend handles requests.
–Added JWT-based login and used Chart.js to show financial data visually in the TaxPal project.

•BharatVersity *Apr 2025 – Jul 2025*
AI/ML Intern BITS Hyderabad

–Built a Music Recommendation System using Retrieval-Augmented Generation (RAG).
–Handled data preprocessing, feature engineering and model evaluation.
–Automated parts of the ML workflow and organised the data pipelines, which improved retrieval accuracy.

Projects

•AI Voice Assistant Dec 2025

Built a voice assistant that uses OpenAI and Gemini APIs to understand speech in real time and reply with generated responses.

•TaxPal Personal Finance & Tax Estimator Feb 2026

Built a full-stack app with React.js, Node.js and MySQL where users can track income and expenses and estimate their tax through RESTful APIs. Added JWT-based authentication for secure login and interactive Chart.js dashboards to show spending and tax insights.

•Real-Time Facial Expression Detection Web App Sep 2026

Built and deployed a web app that detects facial expressions live from the camera using React.js, Flask, ONNX Runtime and MediaPipe. Fixed browser issues on Android and iOS, set up GitHub Actions for testing, deployed with Docker, and added health checks to keep the Render instance alive.

–Live Project Link
–GitHub Repository Link

•RouteWise – Smart Journey Planner with Learning Fares Sep 2026

Built and deployed a journey planner for Visakhapatnam using React, TypeScript, FastAPI, Google Maps APIs and Supabase PostgreSQL. It compares buses, autos, cabs, bikes and walking by cost and travel time, and uses a Bayesian fare model that learns from rider fare votes, filters out outliers and gives more weight to recent votes. Also added live GPS navigation with off-route alerts, hand-off to Google/Apple Maps, Docker deployment on Render, and 62 automated tests.

–Live Project Link
–GitHub Repository Link

•StudyMind – AI Study Workspace & Interview Coach Sep 2026

Built and deployed an AI study platform with React.js, Vite, Firebase and the Google Gemini API. It reads PDFs (with in-browser OCR using Tesseract.js) and uses hybrid BM25 + MiniLM retrieval to give answers with page references, along with notes, 2/5/8-mark question banks and quizzes based on the uploaded document. Also built an Interview Studio that asks adaptive follow-up questions through Gemini and does speech recognition on-device using Whisper via Transformers.js. Added an admin dashboard, Firestore security rules, made uploads 10x faster, wrote 160 Vitest tests, and deployed on Render.

–Live Project Link
–GitHub Repository Link

•SortScope – Interactive DSA Visualizer & Code Tracer Oct 2026

Built and deployed a DSA learning tool using JavaScript, Vite, Node.js and WebAssembly. Users can step forward and backward through five sorting algorithms with matching pseudocode, compare them on the same input, and try 70 operations across seven data structures. The Code Lab runs Python (Pyodide) and JavaScript in the browser line by line, showing variables, call stacks and tree/graph diagrams as the code runs. Wrote 29 Vitest and 30 Playwright tests and deployed on Vercel. Currently adding C, C++ and Java support through a rate-limited compiler API on Oracle Cloud.

–Live Project Link
–GitHub Repository Link

Education

•GITAM University *Aug 2023 – Present*

B.Tech in Computer Science / *CGPA: 8.06* Visakhapatnam

Certifications

•Microsoft Azure AI Fundamentals (AI-900) Apr 2024

Score: 781

Leadership

- Core Member Google Developers Group (GDG), GITAM University 2025 – Present